SOP: AI Lead Qualification Workflow (GoHighLevel + n8n)

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1. Purpose

This SOP explains how an inbound lead travels from a GoHighLevel (GHL) form to a cleaned, AI-scored, human-approved and tagged contact. It exists so that anyone can run, test, fix or hand over the workflow without guessing.

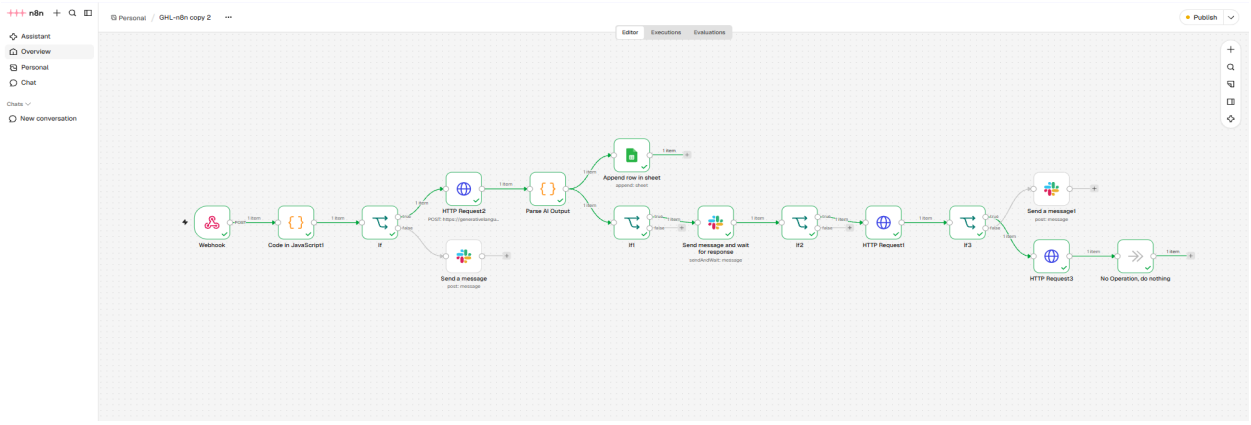
2. Scope and owner

- **Owner:** Olowoyo Oluwaseyi
- **Systems:** GoHighLevel (CRM and forms), n8n (logic), Gemini API (lead scoring), Slack (alerts and approval), Google Sheets (log), Twilio (SMS)
- **Applies to:** test runs and any client build copied from this one

3. How a lead flows

1. A lead submits the GHL form, including the SMS consent checkbox.
2. A GHL workflow fires a Custom Webhook into n8n.
3. n8n cleans and validates the data.
4. Bad leads (missing first name or invalid email) go to a Slack alert and stop.
5. Good leads are sent to the Gemini API, which returns a score, category, reason and draft reply as strict JSON.
6. n8n validates that JSON. If it is malformed, a Slack alert fires and nothing else happens.
7. Every AI call is logged to the AI Lead Log sheet.
8. A Slack message asks a human to approve or disapprove, showing the enquiry, score, reason and draft reply.
9. On approval, the contact is tagged in GHL with ai-hot, ai-warm or ai-cold. On disapproval, a Slack message records the rejection and nothing is tagged.

10. Approved leads then pass an SMS consent gate before any text step.



4. Access and credentials

System	What is used	Where it lives
GHL	Private Integration token with contact read/write and custom-fields read scopes	n8n Header Auth credential, value is Bearer followed by the token. Every GHL request also sends a Version header of 2021-07-28
Gemini	API key from Google AI Studio	n8n Header Auth credential, header name x-goog-api-key
Slack	Slack credential in n8n	Alerts channel
Google Sheets	Google credential in n8n	AI Lead Log spreadsheet

Rules

- Never paste a key or token into a node field. Use credentials.
- Never commit credentials, webhook URLs or IDs to GitHub.
- If a token may have been exposed, rotate it in GHL, then update the n8n credential. Rotating invalidates the old token.

5. Node by node

Node	Purpose
Webhook	Receives the lead from GHL. The test URL listens for one call only. The production URL works once the workflow is activated
Clean Lead (Code)	Trims and lowercases data, normalises the phone number to international format, reads the consent value, builds the issues and warnings lists
IF: is_clean	Sends clean leads onward and bad leads to the Slack alert
Gemini request (HTTP)	Asks the model to score the lead and reply with JSON only. Retry on fail is on
Parse AI Output (Code)	Parses the reply and checks the category, score range and required fields. Sets ai_ok

IF: ai_ok	Valid output continues. Invalid output goes to a Slack alert
Google Sheets (Append Row)	Logs timestamp, contact ID, category, score, tokens and model
Slack: Send and Wait	Human approval with Approve and Disapprove buttons
IF: approved	Approved leads continue. Rejected leads get a Slack message
Tag contact (HTTP)	Adds the ai-category tag in GHL
IF: Has SMS Consent	Only consenting leads reach the SMS step. Others get a Slack skip message

6. Validation rules

Blocks the lead: missing first name, invalid email.

Warns only: missing or invalid phone, missing SMS consent.

Phones are normalised to international format using a default country code set at the top of the Clean Lead node. Change it per client. A lead that fails validation produces a Slack alert containing the contact ID and the list of issues.

7. AI qualification

- **Model:** gemini-2.5-flash, called through the Gemini API
- **Required output:** a JSON object with score (integer 1 to 10), category (hot, warm or cold), reason (short) and suggested_reply (short)
- **Checks:** the parser rejects anything that is not valid JSON, has an unknown category, a score outside 1 to 10, or a missing field
- **Principle:** the model only scores and drafts. It never contacts a lead. A human approves first
- **Tokens:** input and output token counts are logged on every call

8. Human approval

The reviewer reads the enquiry, the score, the reason and the draft reply before clicking. Approve applies the tag while disapprove applies nothing and posts a rejection message. Each approval button works once, on the live waiting run.

9. Error handling

Status	Meaning	What the system does
401	The server does not know who is calling (bad or expired token)	Do not retry. Alert, then fix or rotate the credential
403	Known caller, missing permission (scope)	Do not retry. Alert, then add the scope
400	Request rejected before anything happened (bad body or wrong value type)	Do not retry. Log the body and error, then fix the mapping

429	Sending faster than the limit allows	Slow down with batching and retry with a wait. Alert only if retries run out
503	Temporary server overload (seen on the Gemini API)	Retry with a wait. Alert if retries run out

10. Messaging compliance

1. **What registration is.** US A2P 10DLC registration has two parts: brand verification (who is sending) and campaign approval (what is sent and how consent is managed). The business that owns the messages registers under its own legal identity. This workflow is built to send SMS, but no campaign has been registered yet.
2. **Why unregistered texts get filtered.** Carriers filter or throttle unregistered traffic and treat registered traffic as legitimate. The aim is that all traffic is verified and consensual. Register before any real send.
3. **What STOP must do.** A contact who replies STOP is opted out immediately and receives nothing further. The workflow must treat the resulting send error as an opted-out contact: mark it, do not retry, do not alert as a failure. Every first message names the business and says Reply STOP to opt out.
4. **Proof of consent.** The form shows a separate, unticked checkbox with fixed wording (agreement to receive texts, STOP instruction, message and data rates notice). The workflow captures a consent flag, time and source and gates every SMS on the flag.
5. **Texting people who never opted in.** The answer to a client is no: it creates legal risk and hurts deliverability. The compliant route is an opt-in form like this one.

11. Logging and reporting

AI Lead Log columns: timestamp, contact_id, category, score, input_tokens, output_tokens, model, status (pending, approved or rejected).

A Looker Studio dashboard reads this sheet: leads per day, score by category, approved versus rejected, token use over time.

Dashboard link: *https://datastudio.google.com/s/lPLY0oVcyCo*

12. Testing checklist

Run these after any change.

Test	Expected result
Clean lead, approve	Tag applied in GHL, log row written
Clean lead, disapprove	No tag, rejection message in Slack, log row written
Invalid email	Slack alert, no AI call
Broken AI output	Slack alert, no approval message
Consent ticked	Reaches the SMS step
Consent unticked	Slack skip message, SMS step not reached

13. Known limitations

- No real SMS is sent. The Twilio step is a placeholder, and no A2P campaign has been registered.
- The Gemini free tier has daily and per-minute limits and occasional overload errors.

14. Maintenance

- **Rotate the GHL token:** create or rotate it in Private Integrations, then update the n8n credential once. All GHL nodes share it.

- **Change the model:** update the model name in the request URL and check the parser still reads the response fields.
- **New client:** duplicate the workflow, change the default country code, form field names, Slack channel and sheet, and re-run the testing checklist.
- **Before sharing any export:** remove webhook URLs, location and sheet IDs, and any key accidentally left in a node.